

DATABASE SYSTEMS

Lab 01: SQL Core Operations & Filtering

Course Code: CSE370

Semester: Spring 2026

Department of Computer Science and Engineering

REFERENCE TABLE: LAB_GRADES

double(3,2) → date

std_id	name	major	section	days_present	project_marks	cgpa	submission_date
s001	Abir	CS	1	10	18.5	3.91	2018-09-15
s002	Nafis	CSE	1	12	20	3.86	2018-08-15
s003	Tasneem	CS	1	8	18	3.57	2018-09-18
s004	Nahid	ECE	2	7	16.5	3.25	2018-08-20
s005	Arafat	CS	2	11	20	4.0	2018-09-13
s006	Tasneem	CSE	1	12	17.5	3.7	2018-08-15
s007	Muhtadi	ECE	1	10	19	3.67	2018-09-16
s008	Farhana	CSE	2	6	15	2.67	2018-08-16
s009	Naima	CSE	2	12	20	3.7	2018-08-14

A. GETTING STARTED

Log in : `mysql -u root -p`

Log out : `Ctrl + c`

```
CREATE DATABASE employee;
USE employee;
```

51F BCPWF UI QBHF TQBDF JT LFQU JOUFOUJPOBMMZ FNQUZ BT SFR

Q1. Write the SQL query to create the table structure for 'Lab_Grades' shown above with proper data types.

```
CREATE TABLE lab(
  st_id char(4),
  name varchar(30),
  major char(3),
  section char(1),
  present int,
  marks double,
  cgpa decimal(3,2),
  sub_date date);
```

Q2. Write the SQL query to insert the first row of data (s001, Abir, CS, 1, 10, 18.5, 3.91, '2018-09-15') into the 'Lab_Grades' table.

INSERT INTO lab VALUES

('s001', 'Abir', 'CS', '1', '10', '18.5', '3.91', '2018-09-15');

s001	Abir	CS	1	10			

← Tuple/Row inserted

☐ To add multiple rows:-

INSERT INTO lab VALUES

(.....),

(.....),

(.....);

→ Insert (এখানে) semi colon

→ select use হয় যখন Question এ চায় — Display/Find/Retrieve/Show

B. [SELECT] QUERIES

Format : **SELECT** **FROM** **WHERE**

↓ ↓ ↓
এ column দেখাবে এ Table Condition

Q3. Write a query to display the entire 'Lab_Grades' table with all columns and records.

SELECT * FROM lab ;

Q4. Write a query to retrieve only the 'name' and 'submission_date' columns for all students.

SELECT name, submission_date FROM lab ;

Q5. Write a query to select the student name and calculate the total marks out of 25 using the formula: $\text{project_marks} + \text{days_present} * 5 / 12$. Display this calculated column as 'total_marks'.

SELECT name, $\text{project_marks} + \text{days_present} * 5 / 12$ AS total_mark FROM lab ;

↪ calculate করে এ value উঠে total_mark এ বসবে

name	total_mark

Q6. Write a query to display the names of all students converted to uppercase letters.

SELECT UPPER(name) FROM lab ;

Q7. Write a query to view all details sorted by 'name' in descending order, and then by 'submission_date' in ascending order.

SELECT * FROM lab ORDER BY name DESC, sub-date ASC ;

Q8. Write a query to select students whose project marks are between 17 and 19 inclusive.

SELECT * FROM lab WHERE project_mark BETWEEN 17 and 19 ;

Q9. Write a query to retrieve the details of students who submitted their project in August 2018 AND whose marks are greater than 18.

**SELECT * FROM lab WHERE (sub-date BETWEEN '2018-08-01'
AND '2018-08-31') AND (project_marks > 18) ;**

PATTERN MATCHING (LIKE OPERATOR)

Q10. Write a query to find details of students whose name starts with 'a' (Pattern: 'a%').

```
SELECT * FROM lab WHERE name LIKE 'a%';
```

Q11. Write a query to find details of students whose name ends with 'a' (Pattern: '%a').

```
SELECT * FROM lab WHERE name LIKE '%a';
```

Q12. Write a query to find details of students whose name contains at least two 'a's anywhere (Pattern: '%a%a%').

```
SELECT * FROM lab WHERE name LIKE '%a%a%';
```

Q13. Write a query to find details of students whose name starts with 'a' and is followed by exactly 4 characters (Pattern: 'a_____').

```
SELECT * FROM lab WHERE name LIKE 'a_____';
```

Find students who are from brac

name	email
a	a@nsu.com
b	b@brac.com

```
SELECT * FROM lab WHERE email LIKE 'brac.com';
```

SELECT major FROM lab;

major
CSE
CS
EEE
CS
CSE

SELECT DISTINCT major FROM lab;

major
CSE
CS
EEE

Repeat গুলি
ব্যক্তি নয়

change column

change Tuple/
Row

C. CHANGING STRUCTURE & DATA (ALTER, UPDATE, DELETE)

Q14. Write a query to add a new column named 'project_title' with data type char(10) to the table.

ALTER TABLE lab ADD project_title char(10);

New column added

				project_title

Q15. Write a query to modify the data type of 'project_title' from char(10) to varchar(50).

ALTER TABLE lab MODIFY COLUMN project_title varchar(50);

Q16. Write a query to drop/delete the column 'project_title' from the table completely.

ALTER TABLE lab DROP COLUMN project_title;

Q17. Write an update query to change Arafat's major to 'CSE'.

UPDATE lab SET major = 'CSE' WHERE name = 'Arafat';

କୋଟ Table ନାମ କୋଟ update ହେଉ condition

Q18. Explain what happens if you accidentally omit the WHERE clause in an UPDATE statement.

Q19. Write a query to delete Naima's record from the table.

DELETE FROM lab WHERE name = 'Naima';

Table name condition

Q20. Write a query to delete all records of students who were present for less than 8 days.

DELETE FROM lab WHERE days-present < 8;

Table name condition